

CSA1417 – COMPILER DESIGN

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LAB EXPERIMENT

1. Develop a lexical Analyzer to identify identifiers, constants, operators using C Program.

AIM:

To develop a lexical analyser to identify identifiers, constants, and operators using C.

ALGORITHM:

- Read the input expression.
- Scan each character one by one.
- If a character is a letter, identify it as an identifier.
- If a character is a digit, identify it as a constant.
- If a character is an operator such as +, -, *, /, or =, identify it as an operator.
- Ignore spaces and other symbols.

CODE:

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
int main()
```

```
{
    char str[100];
    int i = 0;
    printf("Enter an expression: ");
    fgets(str, sizeof(str), stdin);
    while (str[i] != '\0')
    {
        if (isalpha(str[i]))
        {
            printf("%c - Identifier\n", str[i]);
        }
        else if (isdigit(str[i]))
        {
            printf("%c - Constant\n", str[i]);
        }
        else if (str[i] == '+' || str[i] == '-' ||
                str[i] == '*' || str[i] == '/' ||
                str[i] == '=')
        {
            printf("%c - Operator\n", str[i]);
        }
        i++;
    }
    return 0;
}
```

}

OUTPUT:

```
Output

Enter an expression: a=b+10
a - Identifier
= - Operator
b - Identifier
+ - Operator
1 - Constant
0 - Constant
```

RESULT:

Thus, the lexical analyser was successfully developed in C to identify identifiers, constants, and operators from a given input.

2. Develop a lexical Analyzer to identify whether a given line is a comment or not using C.

AIM:

To develop a lexical analyser to identify whether a given line is a comment or not using C.

ALGORITHM:

- Read a line of text.
- Check whether the line starts with //.

- If yes, it is a single-line comment.
- Check whether the line starts with /* and ends with */.
- If yes, it is a multi-line comment.
- Otherwise, it is not a comment.

CODE:

```
#include <stdio.h>

#include <string.h>

int main()
{
    char str[100];
    printf("Enter a line: ");
    fgets(str, sizeof(str), stdin);
    if (str[0] == '/' && str[1] == '/')
    {
        printf("It is a single-line comment.");
    }
    else if (str[0] == '/' && str[1] == '*')
    {
        if (strstr(str, "*/") != NULL)
            printf("It is a multi-line comment.");
        else
            printf("It is an incomplete comment.");
    }
}
```

```
else
{
    printf("It is not a comment.");
}
return 0;
}
```

OUTPUT:



```
Output
Enter a line: //ENTER A COMMENT
It is a single-line comment.
```

RESULT:

Thus, the lexical analyser was successfully developed in C to identify whether the given line is a comment or not.